

Chapter 9

Molecular Geometry and Bonding Theories

1. The basis of the VSEPR model of molecular bonding is D.

A) regions of electron density on an atom will organize themselves so as to maximize s-character

B) regions of electron density in the valence shell of an atom will arrange themselves so as to maximize overlap

C) atomic orbitals of the bonding atoms must overlap for a bond to form

D) electron domains in the valence shell of an atom will arrange themselves so as to minimize repulsions

E) hybrid orbitals will form as necessary to, as closely as possible, achieve spherical symmetry

2. The electron domain and D.

- A) tetrahedral, trigonal planar
- B) trigonal planar, trigonal planar
- C) trigonal pyramidal, linear
- D) tetrahedral, bent**
- E) trigonal pyramidal, seesaw

Bonded atoms	Lone pair	Generic formula	Hybridization	Molecular geometry	Electron geometry
1	0	AX	S	Linear	Linear
2	0	AX ₂	Sp	Linear	Linear
1	1	AXN	Sp	Linear	Linear
3	0	AX ₃	Sp ²	Trigonal planar	Trigonal planar
2	1	AX ₂ N	Sp ²	Bent	Trigonal planar
1	2	AXN ₂	Sp ²	Linear	Trigonal planar
4	0	AX ₄	Sp ³	Tetrahedral	Tetrahedral
3	1	AX ₃ N	Sp ³	Trigonal pyramid	Tetrahedral
2	2	AX ₂ N ₂	Sp ³	Bent	Tetrahedral
1	3	AXN ₃	Sp ³	Linear	Tetrahedral
3	2	AX ₃ N ₂	Sp ³ d	T-shaped	Trigonal bipyramidal

3. Of the following species, D will have bond angles of 120° .

A) PH_3

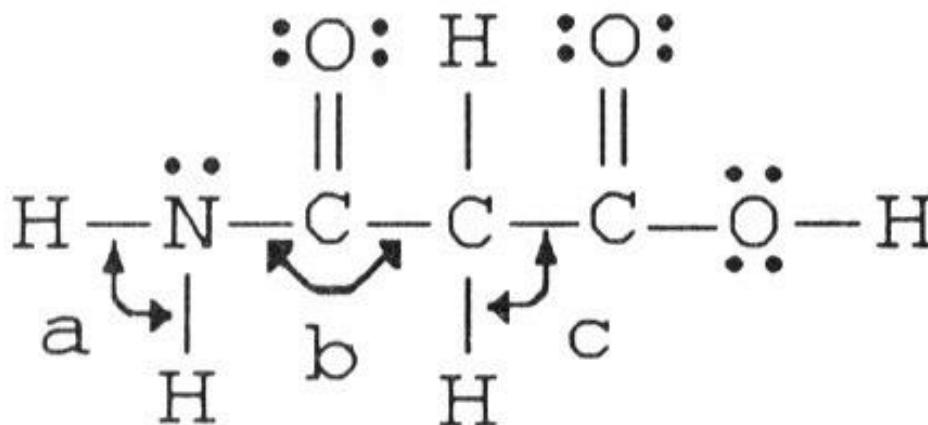
B) ClF_3

C) NCl_3

D) BCl_3 AX_3E_0

E) All of these will have bond angles of 120°

4. The bond angles marked a, b, and c in the molecule below are about _____, _____, and _____ **D** _____, respectively.



- A) 90° , 90° , 90°
- B) 120° , 120° , 90°
- C) 120° , 120° , 109.5°
- D) 109.5° , 120° , 109.5°**
- E) 109.5° , 90° , 120°

5. The central iodine atom in the ICl_4^- ion has _____ nonbonded electron pairs and E bonded electron pairs in its valence shell.

A) 2, 2

B) 3, 4

C) 1, 3

D) 3, 2

E) 2, 4



6. An electron domain consists of D.

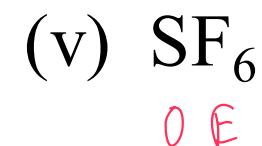
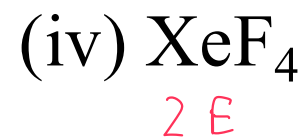
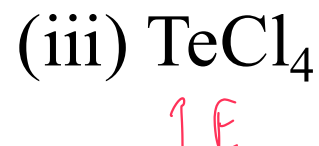
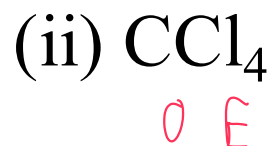
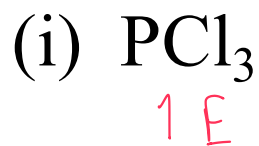
- a) a nonbonding pair of electrons
- b) a single bond
- c) a multiple bond

- A) a only
- B) b only
- C) c only
- D) a, b, and c**
- E) b and c

7. The electron-domain geometry and the molecular geometry of a molecule of the general formula AB_n will always be the same if A.

- A) there are no lone pairs on the central atom
- B) there is more than one central atom
- C) n is greater than four
- D) n is less than four
- E) the octet rule is obeyed

8. For which of the molecules is the molecular geometry (shape) the same as the VSEPR electron domain arrangement (electron domain geometry) C?



- A) (i) and (ii)
- B) (i) and (iii)
- C) (ii) and (v)
- D) (iv) and (v)
- E) (v) only

9. Of the following molecules, only D is polar.



10. An antibonding MO D the corresponding bonding MO.

A) is always lower in energy than

B) can accommodate more electrons than

C) can accommodate fewer electrons than

D) is always higher in energy than

E) is always degenerate with

12. The hybridizations of nitrogen in NF_3 and NH_3 are D and _____, respectively.

A) sp^2 , sp^2

B) sp , sp^3

C) sp^3 , sp

D) sp^3 , sp^3

E) sp^2 , sp^3

13. When four atomic orbitals are mixed to form hybrid orbitals, how many hybrid orbitals are formed D ?

- A) one
- B) six
- C) three
- D) four**
- E) five

14. Valence bond theory does not address the issue of A.

- A) excited states of mol
- B) molecular shape
- C) covalent bonding
- D) hybridization
- E) multiple bonds

This theory focuses on the concepts of **electronic configuration**, atomic orbitals (and their overlapping), and the **hybridization** of these atomic orbitals. Chemical bonds are formed from the overlapping of atomic orbitals wherein the electrons are localized in the corresponding bond region.

The valence bond theory also goes on to explain the electronic structure of the molecules formed by this overlapping of atomic orbitals. It also emphasizes that the nucleus of one atom in a molecule is attracted to the electrons of the other atoms.

Postulates of Valence Bond Theory

The important postulates of the valence bond theory are listed below.

1. **Covalent bonds are formed when two valence orbitals** (half-filled) belonging to two different atoms overlap on each other. The electron density in the area between the two bonding atoms increases as a result of this overlapping, thereby increasing the stability of the resulting molecule.
2. The presence of many unpaired electrons in the valence shell of an atom enables it to form **multiple bonds** with other atoms. The paired electrons present in the valence shell do not take part in the formation of chemical bonds as per the valence bond theory.
3. Covalent chemical bonds are directional and are also **parallel to the region corresponding to the atomic orbitals that are overlapping.**
4. **Sigma bonds** and **pi bonds** differ in the pattern that the atomic orbitals overlap in, i.e. **pi bonds** are formed from sidewise overlapping whereas the overlapping along the axis containing the nuclei of the two atoms leads to the formation of sigma bonds.

15. In molecular orbital theory, the σ_{1s} orbital is _____ and the σ_{1s}^* orbital is **B** _____ in the H_2 molecule.

- A) filled, filled
- B) filled, empty**
- C) filled, half-filled
- D) half-filled, filled
- E) empty, filled

16. Molecular Orbital theory correctly predicts diamagnetism of fluorine gas, F_2 . This is because C.

- A) the bond order in F_2 can be shown to be equal to 1.
- B) there are more electrons in the bonding orbitals than in the antibonding orbitals.
- C) all electrons in the MO electron configuration of F_2 are paired.**
- D) the energy of the π_{2p} MOs is higher than that of the σ_{2p} MO
- E) the F-F bond enthalpy is very low